

Brain Slice Keeper

BSK1

Pre-Incubation of Slice Preparations

The Brain Slice Keeper has been designed to pre-incubate brain slices prior to transfer into recording chambers. It consists of a pair of closely fitting acrylic rings, upper and lower, between which is wedged a sheet of nylon netting.

FEATURES

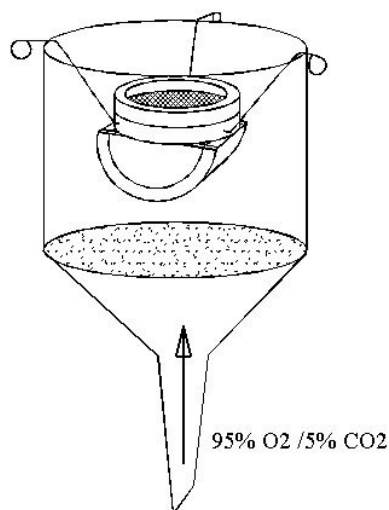
- * Slices maintained for many hours prior to transfer to recording chamber
- * Hooks into purpose designed holding vessel (BSKH) or standard Buchner funnel for bubbling carbogen into ACSF
- * Slices supported on a quick change nylon net
- * Modular design simple to set up and maintain

In operation, the BSK1 is totally immersed in an ACSF-filled purpose designed holding vessel (BSKH) into which a 95% O₂, 5% CO₂ gas mixture is supplied with a ceramic bubbler. The bubbles rising from the base of the BSKH saturate the ACSF and provide constant circulation of medium to the slices which rest on the nylon net. Since the lower ring of the BSK1 carries a deflector shield, bubbles are prevented from being trapped under the slices and ensure continuous circulation of medium. Slices remain viable for many hours in these conditions. Alternatively a standard Buchner funnel with 40 to 60 micron porosity sintered glass membrane can be used instead of the BSKH. When first placing the BSK1 into the BSKH, any large air pockets are dispelled from under the net by means of a teat pipette. When in place, the fluid level is maintained at least 5mm above the upper ring of the BSK1. This ensures circulation of fluid over the top and downwards towards the slices. The typical fluid volume with BSK1 with a BSKH is 220ml and with a Buchner funnel it is 320ml.

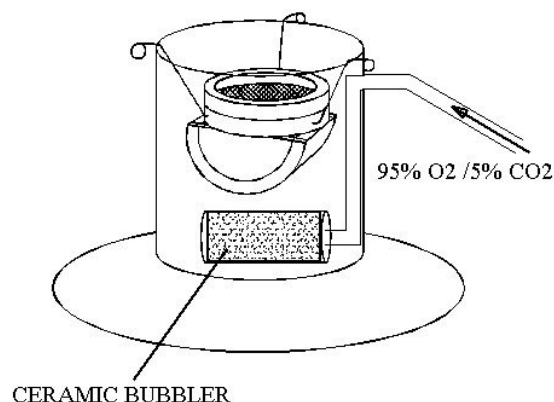


SUPPORT METHODS FOR BRAIN SLICE KEEPER BSK1

BUCHNER FUNNEL



BSKH HOLDER FOR BSK1



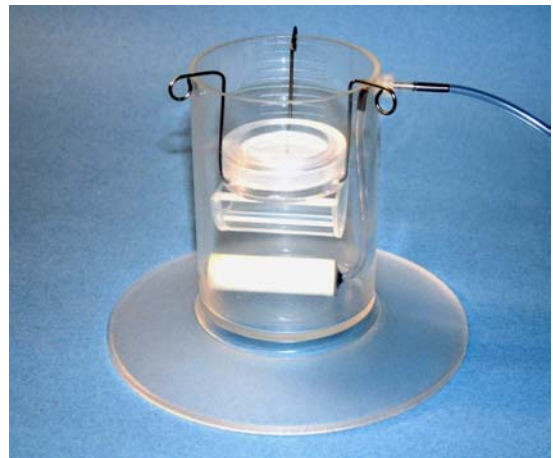
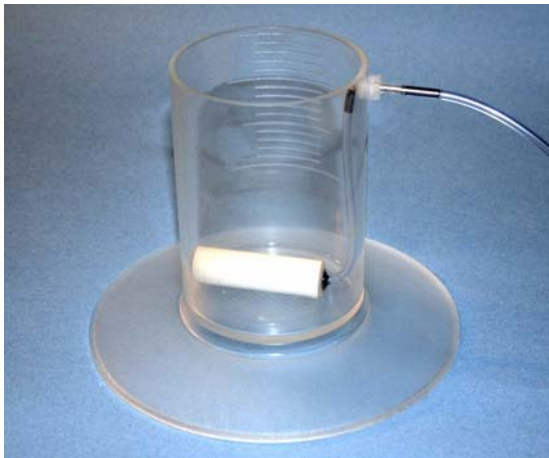
CERAMIC BUBBLER

Brain Slice Keeper

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Pre-Incubation of Slice Preparations

Brain Slice Keeper arrangement in its holder (BSKH)



Schematic arrangement of BSK1 with Buchner funnel and approximate dimensions of BSK1

